R20

Code No.: 20CSC21

CHAITANYA BHARATHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.E. (CSE) V Sem (Main) Examination Dec 2022 / Jan 2023

Data Communication and Computer Networks

Time: 3 Hours **Max Marks: 60**

Note: Answer **ALL** questions from **Part-A & Part –B (Internal Choice)** at one place in the same order

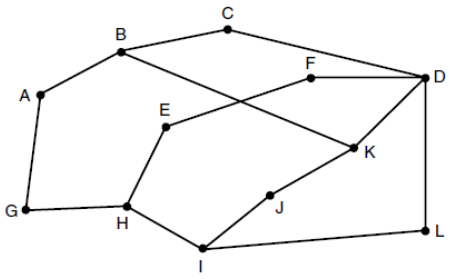
Part - A
(5Q X 3M = 15 Marks)

			M	CO	BT
1		Name the four basic network topologies, and cite an advantage of each type.	(3)	1	1
2		What is the relationship between a switch and a bridge?	(3)	2	2
3		Explain the difference between classful addressing and classless addressing in IPv4?	(3)	3	2
4		What is the maximum and minimum size of the process data that can be encapsulated in a UDP datagram?	(3)	4	1
5		In electronic mail, explain the tasks of a user agent.	(3)	5	2

Part – B
(5Q X 9M = 45 Marks)

			M	CO	BT
6	(a)	When a party makes a local telephone call to another party, is this a point-to-point or multipoint connection? Explain your answer.	(5)	1	2
	(b)	Illustrate the ISO-OSI Reference model. Explain the Critique of the OSI Model and Protocols.	(4)	1	2
		(OR)			
7	(a)	Explain the two major categories of transmission media. Write the advantages of optical fiber over twisted-pair and coaxial cable.	(5)	2	2
	(b)	Compare and contrast a circuit-switched network and a packet-switched network. What is the role of the address field in a packet traveling through a virtual-circuit network?	(4)	2	2
8	(a)	A bit stream 10011101 is transmitted using the standard CRC method described in the text. The generator polynomial is $x^3 + 1$. Show the actual bit string transmitted. Suppose that the third bit from the left is inverted during transmission. Show that this error is detected at the receiver's end. Give an example of bit errors in the bit string transmitted that will not be detected by the receiver.	(5)	2	3
	(b)	A system uses the Go-back-N ARQ Protocol with a window size of 7. If each packet carries 1000 bits of data, how long does it take to send 1 million bits of data if the distance between the sender and receiver is 5000 KM and the propagation speed is 2 x 108 m? Ignore transmission, waiting, and processing delays. Also, ignore the overhead due to the header and trailer. We assume no data or control frame is lost or damaged.	(4)	2	3
		(OR)			

9	(a)	Six stations, *A* through *F*, communicate using the MACA protocol. Is it possible for two transmissions to take place simultaneously? Explain your answer.	(5)	2	3
	(b)	Compare and contrast HDLC with PPP. Which one is byte-oriented; which one is bit-oriented? Explain your answer.	(4)	2	2
10	(a)	Explain the various design challenges of routing algorithms.	(5)	3	2
	(b)	Compute a multicast spanning tree for router *C* in the following network for a group with members at routers *A*, *B*, *C*, *D*, *E*, *F*, *I*, and *K*.	(4)	3	3



(OR)

11	(a)	Illustrate the Message format and the header of IPv6 (fixed header).	(5)	4	2
	(b)	Explain the working of the Internet Control Message Protocol (IMCP).	(4)	4	2
12	(a)	Compare the TCP header and the UDP header. List the fields in the TCP header that are missing from UDP header. Give the reason for their absence.	(5)	5	2
	(b)	Datagram fragmentation and reassembly are handled by IP and are invisible to TCP. Does this mean that TCP does not have to worry about data arriving in the wrong order? Explain your answer.	(4)	5	2

(OR)

13	(a)	Imagine that a two-way handshake rather than a three-way handshake was used to set up connections. In other words, the third message was not required. Are deadlocks now possible? Explain with one example or show that none exist.	(5)	5	2
	(b)	What is the difference between open-loop congestion control and closed-loop congestion control? Name the policies that can prevent congestion.	(4)	5	1
14	(a)	What are the two main categories of DNS messages? Explain with example.	(5)	6	1
	(b)	Why is a connection establishment for mail transfer needed if TCP has already established a connection? Explain your answer.	(4)	6	2

(OR)

15	(a)	In symmetric-key cryptography, if every person in a group of 10 people needs to communicate with every other person in another group of 10 people, identify how many secret keys are needed?	(5)	6	3
	(b)	How does PGP create a set of security parameters? Explain.	(4)	6	2

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